

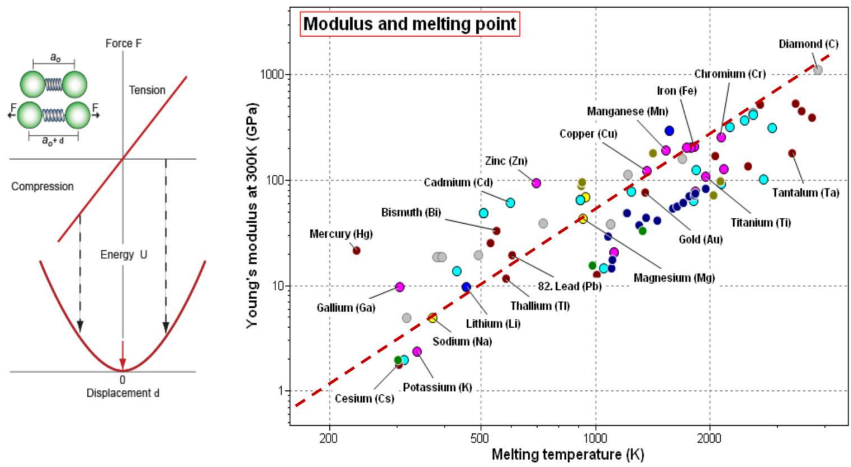
# Material and element characteristics

23 May 2025 21:02

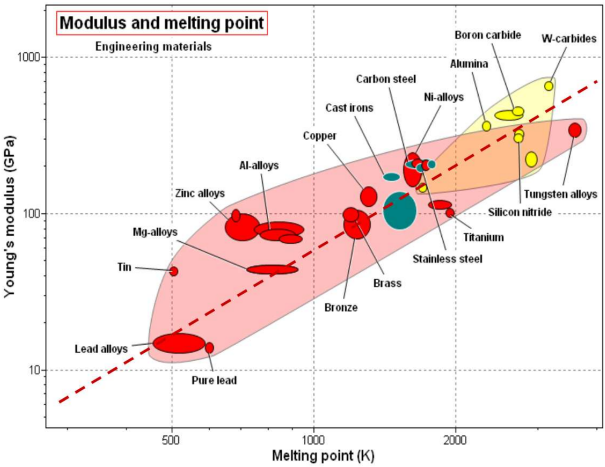
## Property relationships

### ★ Modulus and melting point

#### Elements

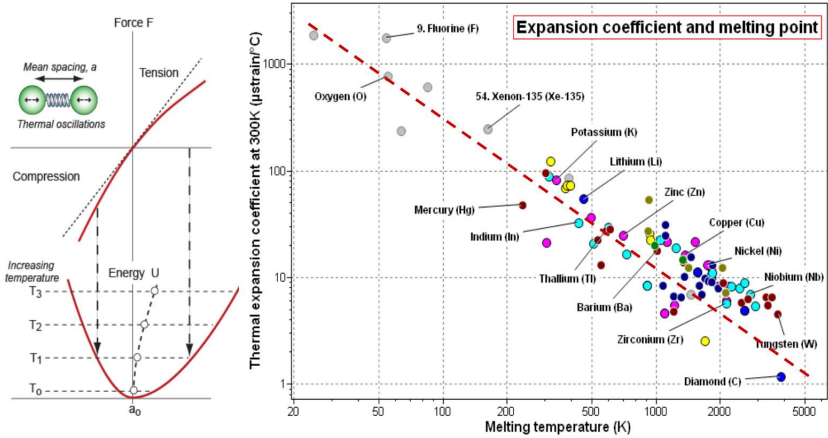


#### Engineering materials

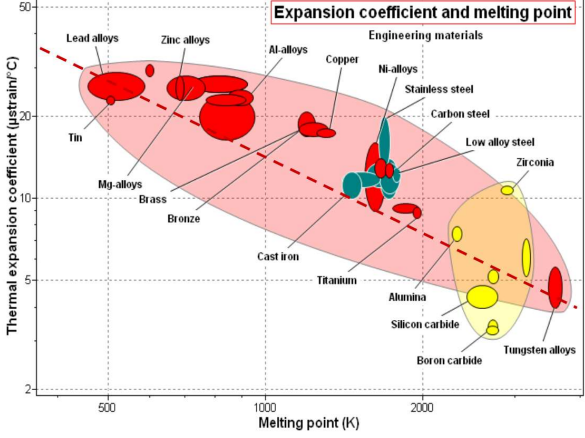


### ★ Expansion coefficient and melting point

#### Elements

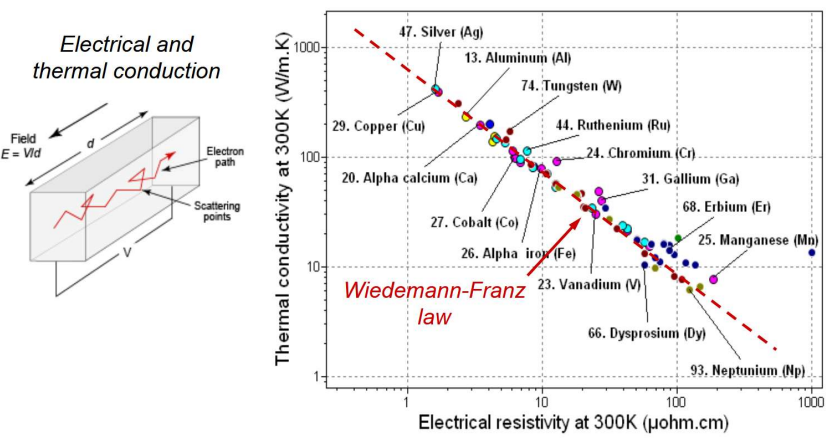


#### Engineering materials

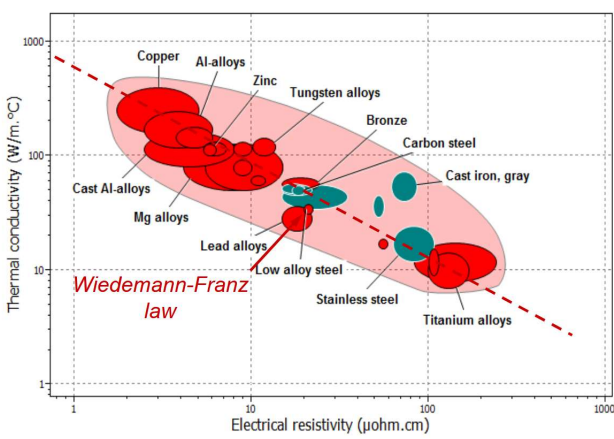


### ★ Electric and thermal conductivities

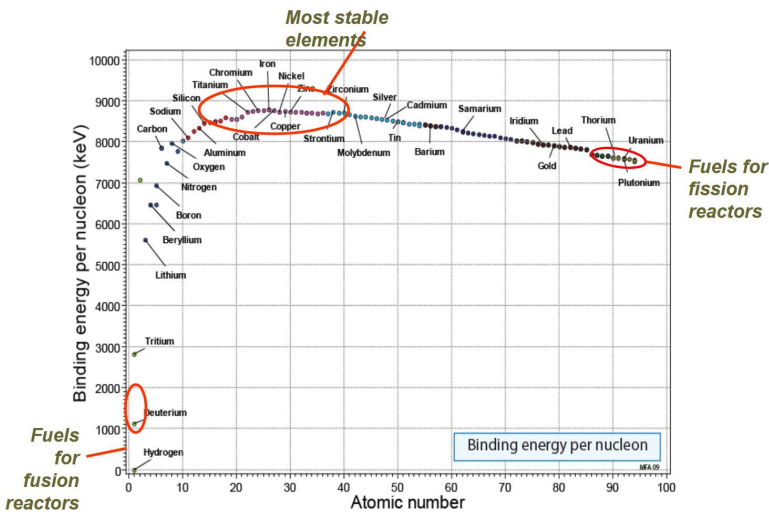
#### Elements



#### Engineering materials



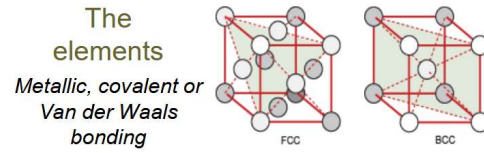
### ★ Binding energy per nucleon



## ★ What determines modulus

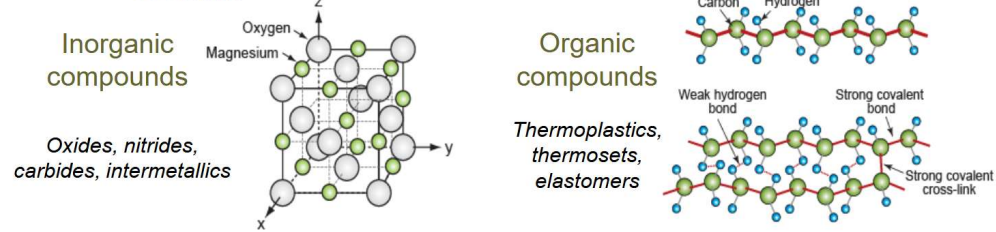
**Density:** atomic weight, atom size and packing density

**Modulus:** interatomic bonds and packing density



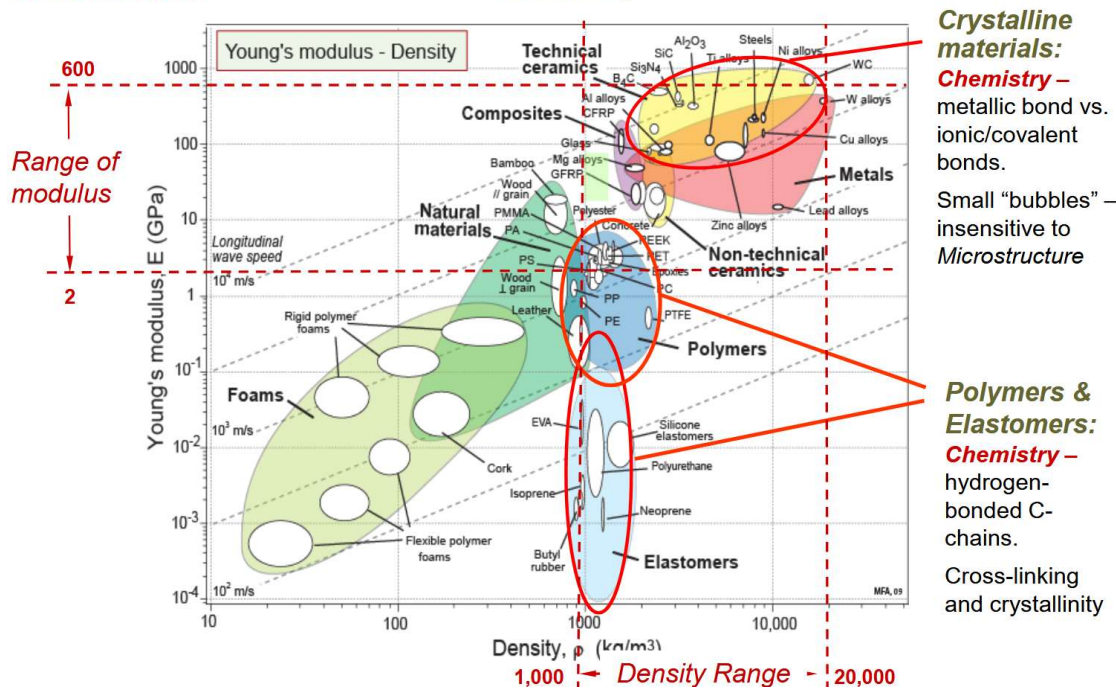
## How manipulate the modulus? - Chemistry

(a) Change the nature of the bonding – ionic, hydrogen, covalent

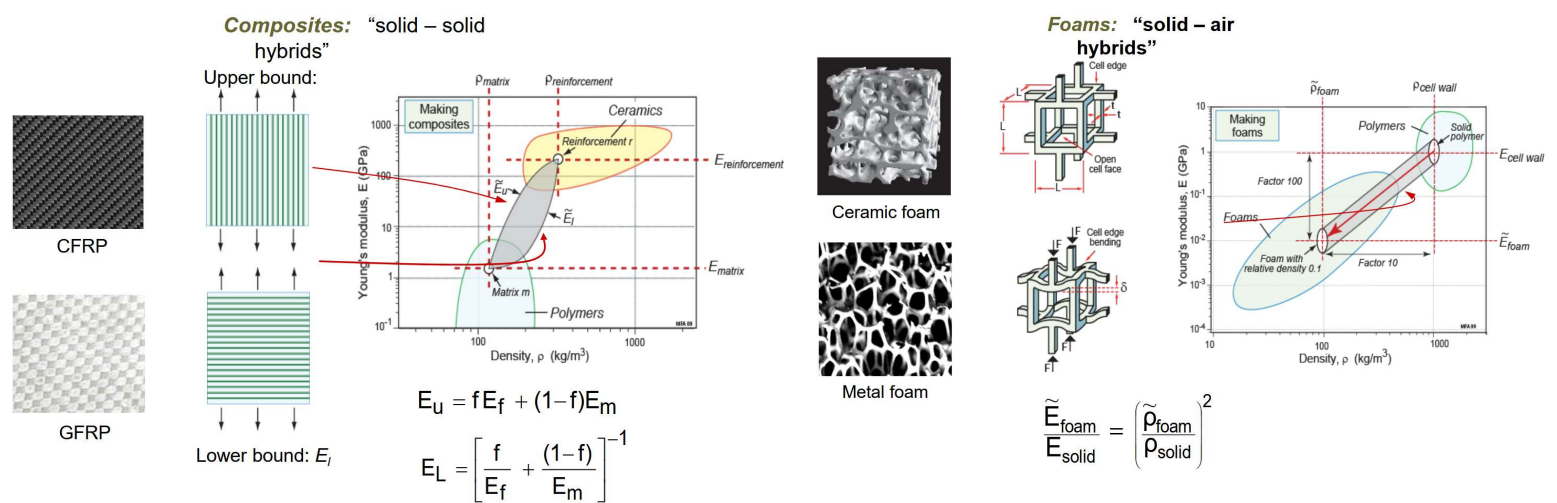


## The elements

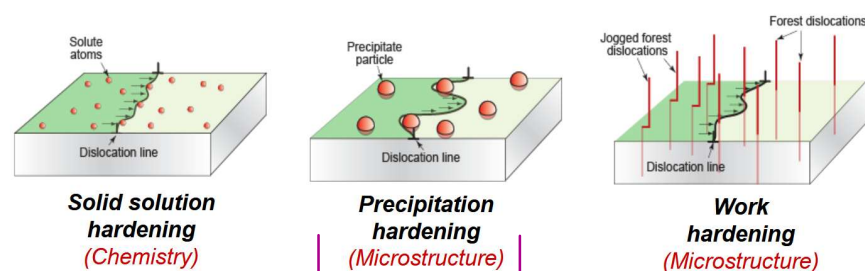
## Chemistry



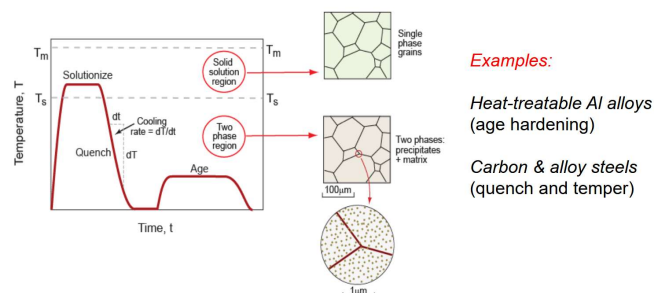
## ★ Control of modulus by architecture



**Metals:** hardening mechanisms – dislocation blocking

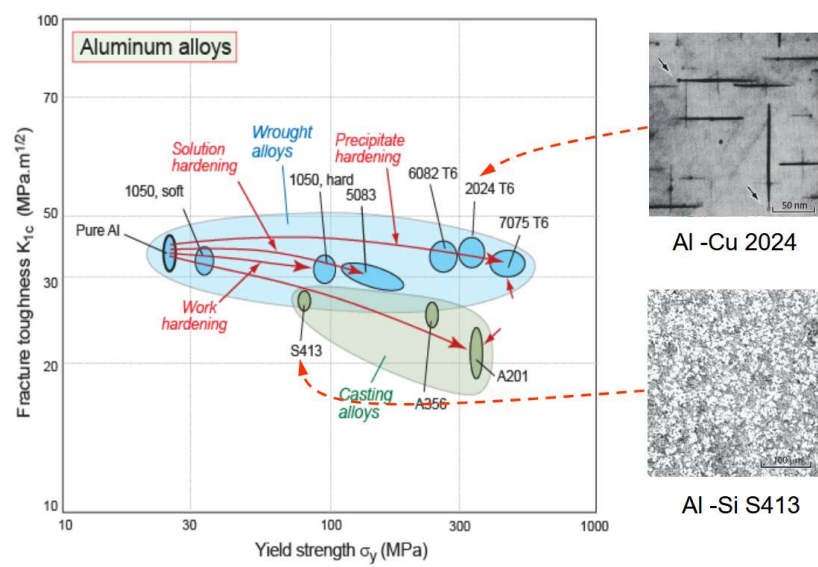


**Metals:** Precipitation hardening

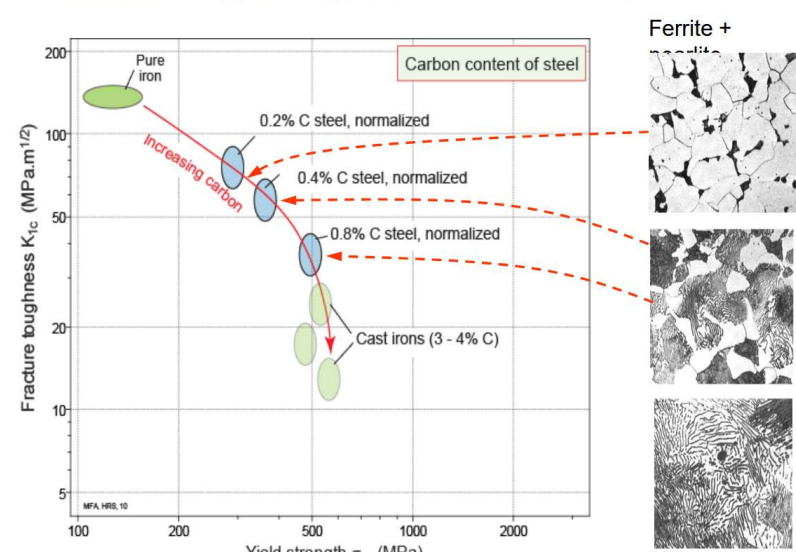




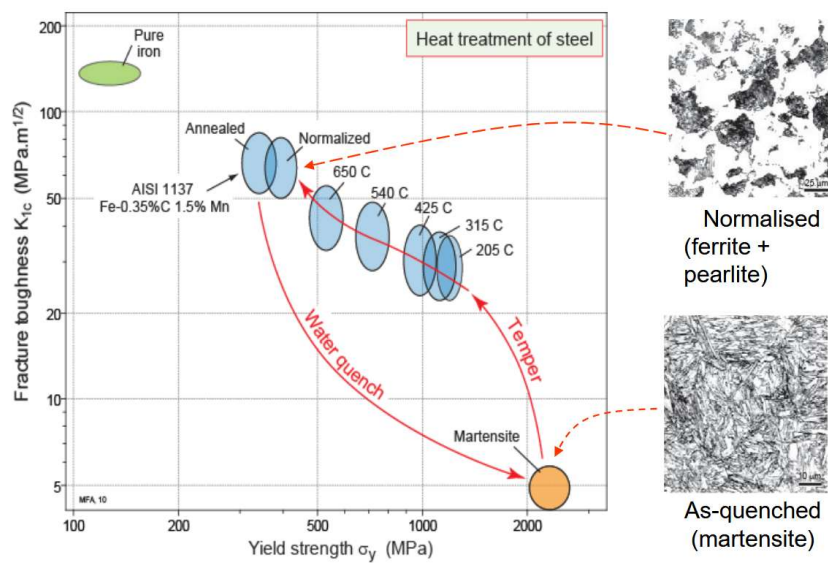
Aluminum alloys: precipitation, solution and work hardening



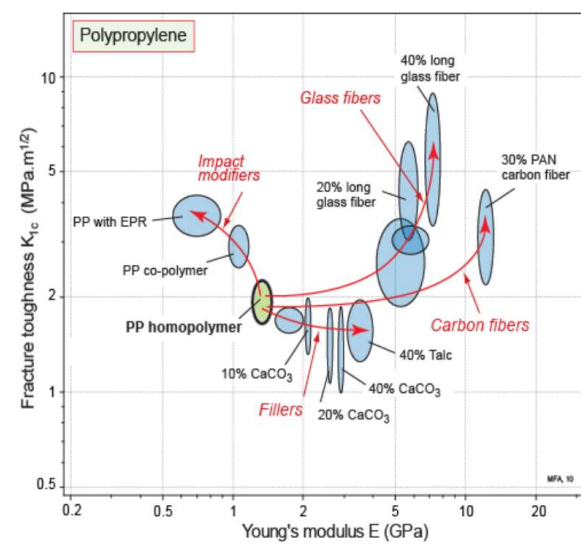
Steels: strength, toughness and carbon content



Steels: Change of microstructure at constant composition



Polymers (PP): fracture toughness-modulus composition trajectory



Chemistry, microstructure and architecture

